PGCP-AC AUGUST 26
Assignment No- 4

Write Create a Employee ,Department & Project Tables .Add atleast 5 records in each table & Write a following Queries.

1. Write a Relational Algebra expression to perform **Cartesian Product (Cross Join)** between Emp and Dept tables.

```
CREATE DATABASE CompanyDB;

USE CompanyDB;

CREATE TABLE Dept (

   DeptID INT PRIMARY KEY,

   DeptName VARCHAR(50),

   Location VARCHAR(50)

);

INSERT INTO Dept VALUES

(1, 'IT', 'Mumbai'),

(2, 'HR', 'Pune'),

(3, 'Finance', 'Mumbai'),

(4, 'Sales', 'Nagpur'),

(5, 'Marketing', 'Pune');

SELECT * FROM Dept;

CREATE TABLE Emp (

   EmpID INT PRIMARY KEY AUTO_INCREMENT,

   EmpName VARCHAR(50),
```

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```
    Salary DECIMAL(10,2),
    DeptID INT,
    FOREIGN KEY (DeptID) REFERENCES Dept(DeptID)
);
INSERT INTO Emp (EmpName, Salary, DeptID) VALUES
('Mansi', 35000, 1),
('Snehal', 40000, 2),
('Jayesh', 45000, 1),
('Khushi', 30000, 3),
('Raj', 50000, 4);
SELECT * FROM Emp;
CREATE TABLE Project (
    ProjectID INT PRIMARY KEY,
    ProjectName VARCHAR(50),
    DeptID INT
);
INSERT INTO Project VALUES
(101, 'Website Development', 1),
(102, 'Recruitment System', 2),
(103, 'Banking System', 3),
(104, 'Sales Management', 4),
(105, 'Marketing Portal', 5);
SELECT * FROM Project;
SHOW TABLES;
```

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```
SELECT * FROM Emp;

SELECT * FROM Dept;

SELECT * FROM Project;
```

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```
mysql> CREATE DATABASE CompanyDB;
Query OK, 1 row affected (0.05 sec)

mysql>
mysql> USE CompanyDB;
Database changed
mysql> CREATE TABLE Dept (
    ->     DeptID INT PRIMARY KEY,
    ->     DeptName VARCHAR(50),
    ->     Location VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.07 sec)

mysql> INSERT INTO Dept VALUES
    -> (1, 'IT', 'Mumbai'),
    -> (2, 'HR', 'Pune'),
    -> (3, 'Finance', 'Mumbai'),
    -> (4, 'Sales', 'Nagpur'),
    -> (5, 'Marketing', 'Pune');
Query OK, 5 rows affected (0.02 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM Dept;
+--------+-----------+----------+
| DeptID | DeptName  | Location |
+--------+-----------+----------+
|      1 | IT        | Mumbai   |
|      2 | HR        | Pune     |
|      3 | Finance   | Mumbai   |
|      4 | Sales     | Nagpur   |
|      5 | Marketing | Pune     |
+--------+-----------+----------+
5 rows in set (0.01 sec)

mysql> CREATE TABLE Emp (
    ->     EmpID INT PRIMARY KEY AUTO_INCREMENT,
    ->     EmpName VARCHAR(50),
    ->     Salary DECIMAL(10,2),
    ->     DeptID INT,
    ->     FOREIGN KEY (DeptID) REFERENCES Dept(DeptID)
    -> );
```

```
Query OK, 0 rows affected (0.03 sec)

mysql> INSERT INTO Emp (EmpName, Salary, DeptID) VALUES
    -> ('Mansi', 35000, 1),
    -> ('Snehal', 40000, 2),
    -> ('Jayesh', 45000, 1),
    -> ('Khushi', 30000, 3),
    -> ('Raj', 50000, 4);
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM Emp;
+-------+---------+----------+--------+
| EmpID | EmpName | Salary   | DeptID |
+-------+---------+----------+--------+
|     1 | Mansi   | 35000.00 |      1 |
|     2 | Snehal  | 40000.00 |      2 |
|     3 | Jayesh  | 45000.00 |      1 |
|     4 | Khushi  | 30000.00 |      3 |
|     5 | Raj     | 50000.00 |      4 |
+-------+---------+----------+--------+
5 rows in set (0.01 sec)

mysql> CREATE TABLE Project (
    ->     ProjectID INT PRIMARY KEY,
    ->     ProjectName VARCHAR(50),
    ->     DeptID INT
    -> );
Query OK, 0 rows affected (0.01 sec)

mysql> INSERT INTO Project VALUES
    -> (101, 'Website Development', 1),
    -> (102, 'Recruitment System', 2),
    -> (103, 'Banking System', 3),
    -> (104, 'Sales Management', 4),
    -> (105, 'Marketing Portal', 5);
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM Project;
```

```
mysql> SELECT * FROM Project;
+-----------+--------------------+--------+
| ProjectID | ProjectName        | DeptID |
+-----------+--------------------+--------+
|       101 | Website Development |      1 |
|       102 | Recruitment System  |      2 |
|       103 | Banking System      |      3 |
|       104 | Sales Management    |      4 |
|       105 | Marketing Portal    |      5 |
+-----------+--------------------+--------+
5 rows in set (0.00 sec)

mysql> SHOW TABLES;
+--------------------+
| Tables_in_companydb |
+--------------------+
| dept               |
| emp                |
| project            |
+--------------------+
3 rows in set (0.02 sec)

mysql>
mysql> SELECT * FROM Emp;
+-------+---------+----------+--------+
| EmpID | EmpName | Salary   | DeptID |
+-------+---------+----------+--------+
|     1 | Mansi   | 35000.00 |      1 |
|     2 | Snehal  | 40000.00 |      2 |
|     3 | Jayesh  | 45000.00 |      1 |
|     4 | Khushi  | 30000.00 |      3 |
|     5 | Raj     | 50000.00 |      4 |
+-------+---------+----------+--------+
5 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM Dept;
+--------+-----------+----------+
| DeptID | DeptName  | Location |
+--------+-----------+----------+
|      1 | IT        | Mumbai   |
|      2 | HR        | Pune     |
|      3 | Finance   | Mumbai   |
|      4 | Sales     | Nagpur   |
|      5 | Marketing | Pune     |
+--------+-----------+----------+
5 rows in set (0.00 sec)
```

2. Write a SQL query to perform an **INNER JOIN** between Emp and Dept tables displaying EmpName and DeptName.

   SELECT Emp.EmpName, Dept.DeptName

   FROM Emp

   INNER JOIN Dept

   ON Emp.DeptID = Dept.DeptID;

```
mysql> SELECT Emp.EmpName, Dept.DeptName
    -> FROM Emp
    -> INNER JOIN Dept
    -> ON Emp.DeptID = Dept.DeptID;
+---------+----------+
| EmpName | DeptName |
+---------+----------+
| Mansi   | IT       |
| Snehal  | HR       |
| Jayesh  | IT       |
| Khushi  | Finance  |
| Raj     | Sales    |
+---------+----------+
5 rows in set (0.00 sec)
```

3. Write a SQL query to perform a **LEFT OUTER JOIN** between Emp and Dept tables to display all employees along with their department information, including employees without a department.

   SELECT Emp.EmpID, Emp.EmpName, Dept.DeptName, Dept.Location

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FROM Emp

LEFT OUTER JOIN Dept

ON Emp.DeptID = Dept.DeptID;

```
mysql> SELECT Emp.EmpID, Emp.EmpName, Dept.DeptName, Dept.Location
    -> FROM Emp
    -> LEFT OUTER JOIN Dept
    -> ON Emp.DeptID = Dept.DeptID;
+-------+---------+----------+----------+
| EmpID | EmpName | DeptName | Location |
+-------+---------+----------+----------+
|     1 | Mansi   | IT       | Mumbai   |
|     2 | Snehal  | HR       | Pune     |
|     3 | Jayesh  | IT       | Mumbai   |
|     4 | Khushi  | Finance  | Mumbai   |
|     5 | Raj     | Sales    | Nagpur   |
+-------+---------+----------+----------+
5 rows in set (0.01 sec)
```

4. Write a SQL query to perform a **RIGHT OUTER JOIN** between Emp and Dept tables.

SELECT Emp.EmpID, Emp.EmpName, Dept.DeptName, Dept.Location

FROM Emp

RIGHT OUTER JOIN Dept

ON Emp.DeptID = Dept.DeptID;

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```
mysql> SELECT Emp.EmpID, Emp.EmpName, Dept.DeptName, Dept.Location
    -> FROM Emp
    -> RIGHT OUTER JOIN Dept
    -> ON Emp.DeptID = Dept.DeptID;
+-------+---------+-----------+----------+
| EmpID | EmpName | DeptName  | Location |
+-------+---------+-----------+----------+
|     1 | Mansi   | IT        | Mumbai   |
|     3 | Jayesh  | IT        | Mumbai   |
|     2 | Snehal  | HR        | Pune     |
|     4 | Khushi  | Finance   | Mumbai   |
|     5 | Raj     | Sales     | Nagpur   |
|  NULL | NULL    | Marketing | Pune     |
+-------+---------+-----------+----------+
6 rows in set (0.00 sec)
```

5. Write a SQL query to perform a **NATURAL JOIN** between Emp and Dept.

SELECT *

FROM Emp

NATURAL JOIN Dept;

```
mysql> SELECT *
    -> FROM Emp
    -> NATURAL JOIN Dept;
+--------+-------+---------+----------+----------+----------+
| DeptID | EmpID | EmpName | Salary   | DeptName | Location |
+--------+-------+---------+----------+----------+----------+
|      1 |     1 | Mansi   | 35000.00 | IT       | Mumbai   |
|      2 |     2 | Snehal  | 40000.00 | HR       | Pune     |
|      1 |     3 | Jayesh  | 45000.00 | IT       | Mumbai   |
|      3 |     4 | Khushi  | 30000.00 | Finance  | Mumbai   |
|      4 |     5 | Raj     | 50000.00 | Sales    | Nagpur   |
+--------+-------+---------+----------+----------+----------+
5 rows in set (0.01 sec)
```

6. Write a SQL query to perform a **CROSS JOIN** between Emp and Project table.

SELECT Emp.EmpName, Project.ProjectName

FROM Emp

CROSS JOIN Project;

```
mysql> SELECT *
    -> FROM Emp
    -> NATURAL JOIN Dept;
+--------+-------+---------+----------+----------+----------+
| DeptID | EmpID | EmpName | Salary   | DeptName | Location |
+--------+-------+---------+----------+----------+----------+
|      1 |     1 | Mansi   | 35000.00 | IT       | Mumbai   |
|      2 |     2 | Snehal  | 40000.00 | HR       | Pune     |
|      1 |     3 | Jayesh  | 45000.00 | IT       | Mumbai   |
|      3 |     4 | Khushi  | 30000.00 | Finance  | Mumbai   |
|      4 |     5 | Raj     | 50000.00 | Sales    | Nagpur   |
+--------+-------+---------+----------+----------+----------+
5 rows in set (0.01 sec)
```

SELECT Emp.EmpName, Project.ProjectName

FROM Emp

CROSS JOIN Project;

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```
mysql> SELECT Emp.EmpName, Project.ProjectName
    -> FROM Emp
    -> CROSS JOIN Project;
+---------+---------------------+
| EmpName | ProjectName         |
+---------+---------------------+
| Raj     | Website Development |
| Khushi  | Website Development |
| Jayesh  | Website Development |
| Snehal  | Website Development |
| Mansi   | Website Development |
| Raj     | Recruitment System  |
| Khushi  | Recruitment System  |
| Jayesh  | Recruitment System  |
| Snehal  | Recruitment System  |
| Mansi   | Recruitment System  |
| Raj     | Banking System      |
| Khushi  | Banking System      |
| Jayesh  | Banking System      |
| Snehal  | Banking System      |
| Mansi   | Banking System      |
| Raj     | Sales Management    |
| Khushi  | Sales Management    |
| Jayesh  | Sales Management    |
| Snehal  | Sales Management    |
| Mansi   | Sales Management    |
| Raj     | Marketing Portal    |
| Khushi  | Marketing Portal    |
| Jayesh  | Marketing Portal    |
| Snehal  | Marketing Portal    |
| Mansi   | Marketing Portal    |
+---------+---------------------+
25 rows in set (0.01 sec)
```

7. Write a SQL query to **create a new table EmpBackup with the same structure as Emp but no data**.

CREATE TABLE EmpBackup LIKE Emp;

SELECT * FROM EmpBackup;

DESCRIBE EmpBackup;

CREATE TABLE EmpBackup LIKE Emp;

SELECT * FROM EmpBackup;

DESCRIBE EmpBackup;

```
mysql> CREATE TABLE EmpBackup LIKE Emp;
Query OK, 0 rows affected (0.03 sec)

mysql> SELECT * FROM EmpBackup;
Empty set (0.01 sec)

mysql> DESCRIBE EmpBackup;
+---------+--------------+------+-----+---------+----------------+
| Field   | Type         | Null | Key | Default | Extra          |
+---------+--------------+------+-----+---------+----------------+
| EmpID   | int          | NO   | PRI | NULL    | auto_increment |
| EmpName | varchar(50)  | YES  |     | NULL    |                |
| Salary  | decimal(10,2)| YES  |     | NULL    |                |
| DeptID  | int          | YES  | MUL | NULL    |                |
+---------+--------------+------+-----+---------+----------------+
4 rows in set (0.02 sec)

mysql> CREATE TABLE EmpBackup LIKE Emp;
ERROR 1050 (42S01): Table 'empbackup' already exists
mysql>
mysql> SELECT * FROM EmpBackup;
Empty set (0.00 sec)

mysql>
mysql> DESCRIBE EmpBackup;
+---------+--------------+------+-----+---------+----------------+
| Field   | Type         | Null | Key | Default | Extra          |
+---------+--------------+------+-----+---------+----------------+
| EmpID   | int          | NO   | PRI | NULL    | auto_increment |
| EmpName | varchar(50)  | YES  |     | NULL    |                |
| Salary  | decimal(10,2)| YES  |     | NULL    |                |
| DeptID  | int          | YES  | MUL | NULL    |                |
+---------+--------------+------+-----+---------+----------------+
4 rows in set (0.00 sec)
```

8. Write a SQL query to **copy all data from Emp into EmpBackup table**.

INSERT INTO EmpBackup

SELECT * FROM Emp;

SELECT * FROM EmpBackup;

```
mysql> INSERT INTO EmpBackup
    -> SELECT * FROM Emp;
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM EmpBackup;
+-------+---------+----------+--------+
| EmpID | EmpName | Salary   | DeptID |
+-------+---------+----------+--------+
|     1 | Mansi   | 35000.00 |      1 |
|     2 | Snehal  | 40000.00 |      2 |
|     3 | Jayesh  | 45000.00 |      1 |
|     4 | Khushi  | 30000.00 |      3 |
|     5 | Raj     | 50000.00 |      4 |
+-------+---------+----------+--------+
5 rows in set (0.00 sec)
```

9. Write a SQL query to **create a new table ProjectArchive with the same structure and data as Project**.

   CREATE TABLE ProjectArchive AS

   SELECT * FROM Project;

   CREATE TABLE ProjectArchive AS

   SELECT * FROM Project;

   SELECT * FROM ProjectArchive;

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```
mysql> CREATE TABLE ProjectArchive AS
    -> SELECT * FROM Project;
Query OK, 5 rows affected (0.12 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM ProjectArchive;
+-----------+--------------------+--------+
| ProjectID | ProjectName        | DeptID |
+-----------+--------------------+--------+
|       101 | Website Development |      1 |
|       102 | Recruitment System  |      2 |
|       103 | Banking System      |      3 |
|       104 | Sales Management    |      4 |
|       105 | Marketing Portal    |      5 |
+-----------+--------------------+--------+
5 rows in set (0.00 sec)
```

10. Write a SQL query to create an **AUTO_INCREMENT sequence for EmpID in the Emp table** during table creation.

```
CREATE TABLE Emp (
   EmpID INT PRIMARY KEY AUTO_INCREMENT,
   EmpName VARCHAR(50),
   Salary DECIMAL(10,2),
   DeptID INT
);
DESCRIBE Emp;
INSERT INTO Emp (EmpName, Salary, DeptID)
VALUES ('Mansi', 42000, 1);
SELECT * FROM Emp;
```

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```
mysql> CREATE TABLE Emp (
    ->     EmpID INT PRIMARY KEY AUTO_INCREMENT,
    ->     EmpName VARCHAR(50),
    ->     Salary DECIMAL(10,2),
    ->     DeptID INT
    -> );
ERROR 1050 (42S01): Table 'emp' already exists
mysql> DESCRIBE Emp;
+---------+--------------+------+-----+---------+----------------+
| Field   | Type         | Null | Key | Default | Extra          |
+---------+--------------+------+-----+---------+----------------+
| EmpID   | int          | NO   | PRI | NULL    | auto_increment |
| EmpName | varchar(50)  | YES  |     | NULL    |                |
| Salary  | decimal(10,2)| YES  |     | NULL    |                |
| DeptID  | int          | YES  | MUL | NULL    |                |
+---------+--------------+------+-----+---------+----------------+
4 rows in set (0.06 sec)

mysql> INSERT INTO Emp (EmpName, Salary, DeptID)
    -> VALUES ('Mansi', 42000, 1);
Query OK, 1 row affected (0.02 sec)

mysql> SELECT * FROM Emp;
+-------+---------+----------+--------+
| EmpID | EmpName | Salary   | DeptID |
+-------+---------+----------+--------+
|     1 | Mansi   | 35000.00 |      1 |
|     2 | Snehal  | 40000.00 |      2 |
|     3 | Jayesh  | 45000.00 |      1 |
|     4 | Khushi  | 30000.00 |      3 |
|     5 | Raj     | 50000.00 |      4 |
|     6 | Mansi   | 42000.00 |      1 |
+-------+---------+----------+--------+
6 rows in set (0.01 sec)
```

11. Write a SQL query using a **subquery** to find all employees whose salary is greater than the **average salary of all employees**.

```
SELECT *

FROM Emp

WHERE Salary > (

  SELECT AVG(Salary)

  FROM Emp
```

);

```
mysql> SELECT *
    -> FROM Emp
    -> WHERE Salary > (
    ->     SELECT AVG(Salary)
    ->     FROM Emp
    -> );
+-------+---------+----------+--------+
| EmpID | EmpName | Salary   | DeptID |
+-------+---------+----------+--------+
|     3 | Jayesh  | 45000.00 |      1 |
|     5 | Raj     | 50000.00 |      4 |
|     6 | Mansi   | 42000.00 |      1 |
+-------+---------+----------+--------+
3 rows in set (0.01 sec)
```

12. Write a SQL query using a **subquery in WHERE clause** to find employees working in departments located in 'Mumbai'.

```
SELECT *

FROM Emp

WHERE DeptID IN (

  SELECT DeptID

  FROM Dept

  WHERE Location = 'Mumbai'

);
```

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```
mysql> SELECT *
    -> FROM Emp
    -> WHERE DeptID IN (
    ->     SELECT DeptID
    ->     FROM Dept
    ->     WHERE Location = 'Mumbai'
    -> );
+-------+---------+----------+--------+
| EmpID | EmpName | Salary   | DeptID |
+-------+---------+----------+--------+
|     1 | Mansi   | 35000.00 |      1 |
|     3 | Jayesh  | 45000.00 |      1 |
|     6 | Mansi   | 42000.00 |      1 |
|     4 | Khushi  | 30000.00 |      3 |
+-------+---------+----------+--------+
4 rows in set (0.00 sec)
```

13. Write a SQL query to display all departments where the number of employees is greater than **5**, using a subquery.

```
SELECT *
FROM Dept
WHERE DeptID IN (
  SELECT DeptID
  FROM Emp
  GROUP BY DeptID
  HAVING COUNT(*) > 5
);
```

```
mysql> SELECT *
    -> FROM Dept
    -> WHERE DeptID IN (
    ->     SELECT DeptID
    ->     FROM Emp
    ->     GROUP BY DeptID
    ->     HAVING COUNT(*) > 5
    -> );
Empty set (0.01 sec)
```

14. Write a subqueries for find out Max salary of employee.

```
SELECT *
FROM Emp
WHERE Salary = (
  SELECT MAX(Salary)
```

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```
   FROM Emp
);
SELECT MAX(Salary) AS MaximumSalary
FROM Emp;
```

```
mysql> SELECT *
    -> FROM Emp
    -> WHERE Salary = (
    ->     SELECT MAX(Salary)
    ->     FROM Emp
    -> );
+-------+---------+----------+--------+
| EmpID | EmpName | Salary   | DeptID |
+-------+---------+----------+--------+
|     5 | Raj     | 50000.00 |      4 |
+-------+---------+----------+--------+
1 row in set (0.00 sec)

mysql> SELECT MAX(Salary) AS MaximumSalary
    -> FROM Emp;
+---------------+
| MaximumSalary |
+---------------+
|      50000.00 |
+---------------+
1 row in set (0.01 sec)
```

15. Write a self join query for each table.

```
SELECT E1.EmpName AS Employee1,
    E2.EmpName AS Employee2
FROM Emp E1
JOIN Emp E2
ON E1.DeptID = E2.DeptID
WHERE E1.EmpID < E2.EmpID;
SELECT D1.DeptName AS Department1,
    D2.DeptName AS Department2
FROM Dept D1
JOIN Dept D2
ON D1.Location = D2.Location
WHERE D1.DeptID < D2.DeptID;
SELECT P1.ProjectName AS Project1,
    P2.ProjectName AS Project2
FROM Project P1
```

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JOIN Project P2

ON P1.DeptID = P2.DeptID

WHERE P1.ProjectID < P2.ProjectID;

```
mysql> SELECT E1.EmpName AS Employee1,
    ->        E2.EmpName AS Employee2
    -> FROM Emp E1
    -> JOIN Emp E2
    -> ON E1.DeptID = E2.DeptID
    -> WHERE E1.EmpID < E2.EmpID;
+-----------+-----------+
| Employee1 | Employee2 |
+-----------+-----------+
| Mansi     | Jayesh    |
| Mansi     | Mansi     |
| Jayesh    | Mansi     |
+-----------+-----------+
3 rows in set (0.00 sec)

mysql> SELECT D1.DeptName AS Department1,
    ->        D2.DeptName AS Department2
    -> FROM Dept D1
    -> JOIN Dept D2
    -> ON D1.Location = D2.Location
    -> WHERE D1.DeptID < D2.DeptID;
+-------------+-------------+
| Department1 | Department2 |
+-------------+-------------+
| IT          | Finance     |
| HR          | Marketing   |
+-------------+-------------+
2 rows in set (0.01 sec)

mysql> SELECT P1.ProjectName AS Project1,
    ->        P2.ProjectName AS Project2
    -> FROM Project P1
    -> JOIN Project P2
    -> ON P1.DeptID = P2.DeptID
    -> WHERE P1.ProjectID < P2.ProjectID;
Empty set (0.00 sec)
```